Ascon

Specification

The Ascon family is specified in NIST SP 800-232: "Ascon-based Lightweight Cryptography Standards for Constrained Devices: Authenticated Encryption, Hash, and Extendable Output Functions".

The original authenticated encryption (AEAD) and hashing schemes (Hash, XOF) Ascon as submitted to the LWC project are specified in the Journal of Cryptology article, the NIST submission, and the CAESAR submission. Additionally, pseudorandom functions (PRF) and message authentication codes (MAC) are specified in the CT-RSA 2024 paper.

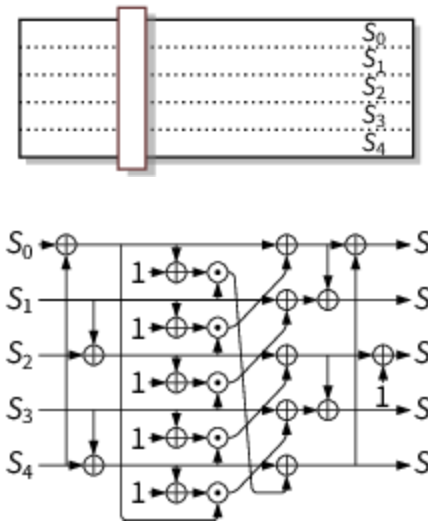
NIST's draft standard is similar, but not identical to the submission. Changes include the selection of a subset of schemes (with new names and IV constants); changing the mode to little-endian bit order; adding a customizable hash variant; and other flexible implementation options. Below, we provide a brief overview of the standardized schemes. For full details including the round constant and IV values, the little-endian bit ordering, padding rules, or the almost identical decryption mode, we refer to NIST SP 800-232.

Ascon's Permutation

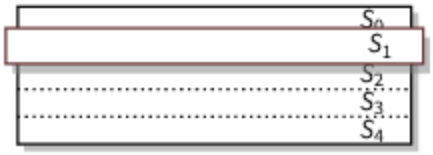
All Ascon family members use the same lightweight permutation. This permutation iteratively applies an SPN-based round transformation for up to 16 rounds; the standardized modes of operation use 12 rounds in Ascon-p[12], and 8 rounds in Ascon-p[8]. The round transformation consists of the following three steps which operate on a 320-bit state divided into 5 words S_0, S_1, S_2, S_3, S_4 of 64 bits each:

- **Addition of Round Constants**: XORs a round-specific 1-byte constant to word S_2.

- **Nonlinear Substitution Layer**: applies a 5-bit S-box 64 times in parallel in a bit-sliced fashion (vertically, across words).
- **Linear Diffusion Layer**: XORs different rotated copies of each word (horizontally, within each word).



ASCON's S-box [tex] [C instructions]



$$S_0 := S_0 \oplus (S_0 \ggg 19) \oplus (S_0 \ggg 28)$$
$$S_1 := S_1 \oplus (S_1 \ggg 61) \oplus (S_1 \ggg 39)$$
$$S_2 := S_2 \oplus (S_2 \ggg 1) \oplus (S_2 \ggg 6)$$
$$S_3 := S_3 \oplus (S_3 \ggg 10) \oplus (S_3 \ggg 17)$$
$$S_4 := S_4 \oplus (S_4 \ggg 7) \oplus (S_4 \ggg 41)$$

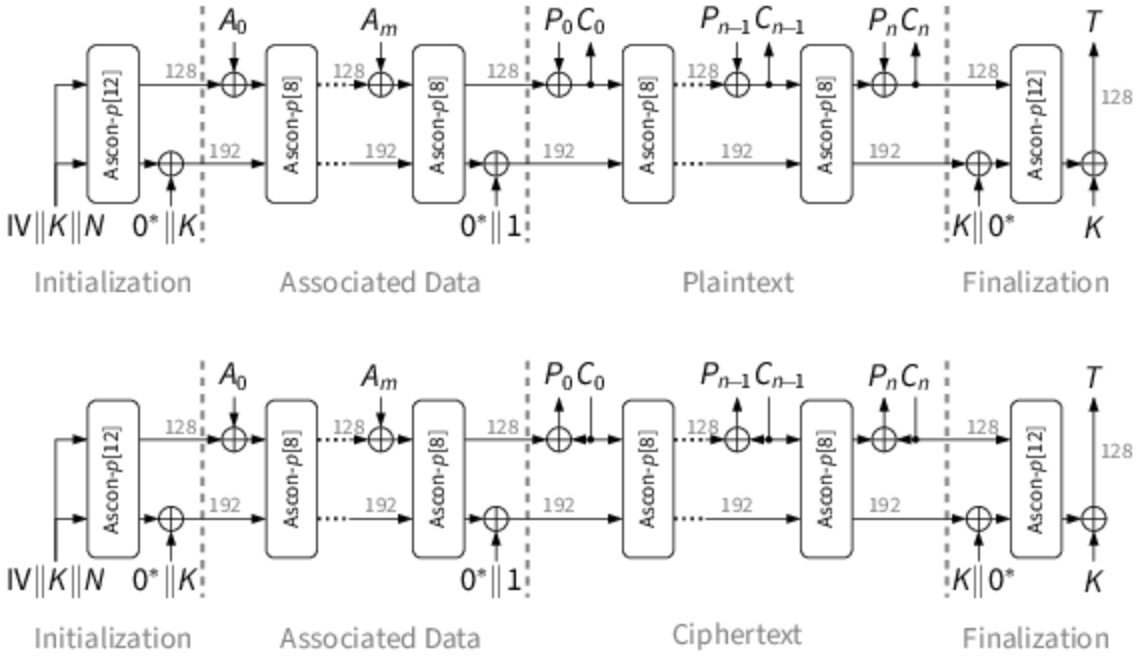
ASCON's linear layer

ASCON's permutation: $\oplus$ denotes XOR, $\odot$ denotes AND, $\ggg$ is rotation to the right.

Authenticated Encryption with ASCON-AEAD128

ASCON uses a duplex-based mode of operation for authenticated encryption. The key, tag and nonce length is 128 bits. The sponge operates on a state of 320 bits, with message blocks of 128 bits. The encryption process is split into four phases:

1. **Initialization**: initializes the state with the key K and nonce N.
2. **Associated Data Processing**: updates the state with associated data blocks A_i.
3. **Plaintext Processing**: injects plaintext blocks P_i into the state and extracts ciphertext blocks C_i.
4. **Finalization**: injects the key K again and extracts the tag T for authentication.



Ascon-AEAD128 authenticated **encryption** (top) and **decryption** (bottom) [tex]

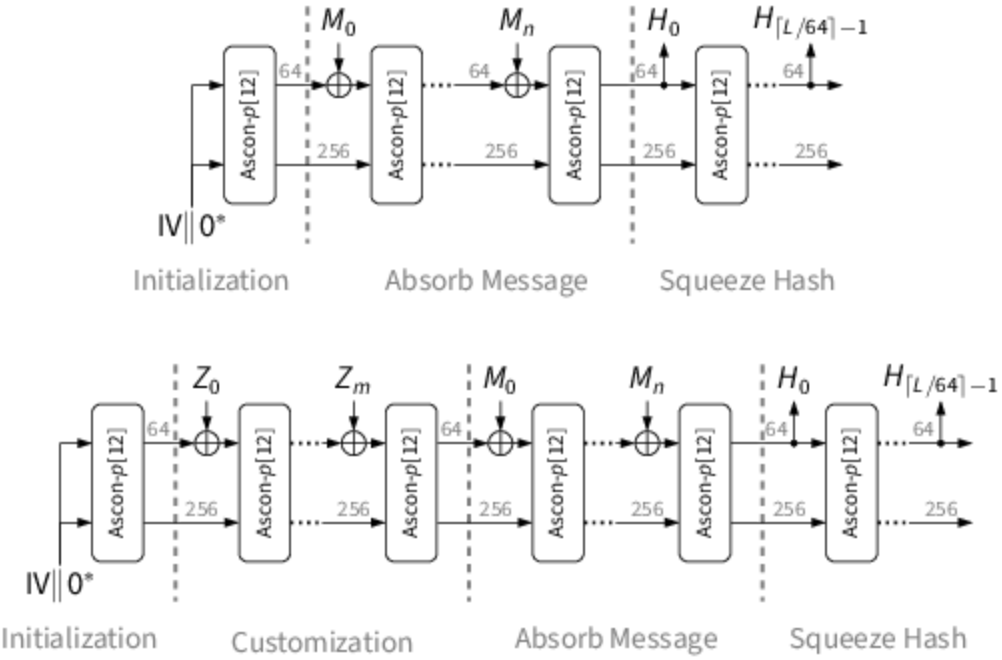
After each injected block (except the last plaintext block), the core permutation Ascon-p[8] is applied to the complete state. During initialization and finalization, a stronger permutation Ascon-p[12] with more rounds is used.

Parameters for Ascon authenticated encryption

Cipher	Bit size of					Rounds	
	key	nonce	tag	rate	capacity	p[12]	p[8]
Ascon-AEAD128	128	128	128	128	192	12	8

Hashing with Ascon-Hash256 and Ascon-(C)XOF128

The Ascon family includes the hash functions Ascon-Hash and the extendable output functions Ascon-XOF128 with its customizable variant Ascon-CXOF128. The hashing modes use the same lightweight 320-bit permutation as the authenticated encryption modes. All variants use similar sponge-based modes of operation and provide 128-bit security when using a hash size of at least 256 bits.



Hashing mode for Ascon-Hash256, Ascon-XOF128 (top) and Ascon-CXOF128 (bottom) [tex]

The hashing modes absorb the message M in 64-bit blocks M_i and finally squeeze the hash value H in 64-bit blocks H_i. The customizable variant additionally absorbs a customization string Z in 64-bit blocks Z_i before the message M. After each absorbed or squeezed block, the permutation Ascon-p[12] is applied to the state.

Parameters for Ascon hashing

Algorithm	Bit size of			Rounds
	hash output	rate	capacity	p[12]
Ascon-Hash256	256	64	256	12
Ascon-XOF128	arbitrary	64	256	12
Ascon-CXOF128	arbitrary	64	256	12